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Palaeomagnetism of Vendian–Early Cambrian sedimentary rocks from E Finnmark, Norway

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Abstract

A palaeomagnetic study has been made on Vendian–Early Cambrian rocks of the Tanafjord and Vestertana groups from the Varanger Peninsula, northernmost Norway. Magnetic components of probably primary origin were isolated at two sites apparently unaffected by the Finnmarkian–Scandian events of the Caledonian Orogeny. Sites situated in the Caledonian Gaissa Nappe or close to its front were remagnetized and comparison with a tentative Ordovician to Devonian apparent polar wander path indicates that the remagnetization occurred during these periods.

1. Introduction

The Fennoscandian Apparent Polar Wander Path (APWP) is at present well defined for parts of the Middle Proterozoic and for the middle to late Palaeozoic (Pesonen et al., 1989; Elming et al., 1993). However, there are significant gaps, e.g., the late Riphean to Late Cambrian interval. This interval is an important period in the Earth's history which appears to have involved the breaking up of large landmasses. Recently, new ideas of the ancient configuration of the continents during these times have emerged (e.g., Hoffman, 1991). Torsvik et al. (1990) have also shown that during the Ordovician the Fennoscandian Shield (Baltica) was situated at southern latitudes and orientated "upside down" in relation to its present position. This makes it important to fill the gap in the APWP mentioned above in order to trace the movements of Fennoscandia during this time interval.

Recently Torsvik and Trench (1991) presented a detailed palaeomagnetic study of Llanvirnian–Caradocian limestones from southern Sweden and resolved a possible primary pole for the Fennoscandian APWP. On the Precambrian side of the gap there are well documented data from the Sveconorwegian part of the Fennoscandian Shield up to ca. 850 Ma. The gap itself is partly filled with data from the Kola Peninsula (Khramov, 1984, 1986, 1989), the late Riphean to early Vendian Vadsø and Barents Sea Group of the Varanger Peninsula (Bylund, in press), the Båtsfjord dykes from the allochthonous part of the Varanger Peninsula (Kjøde et al., 1978; Kjøde, 1980), the Nexö sandstone of Bornholm, Denmark (Prasad and Sharma, 1978) and the alkaline intrusions of Fen (Poorter, 1972; Piper, 1988) and

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Alnön (Piper, 1981). In this paper palaeomagnetic results from the Tanafjord and Vestertana groups of the Varangerfjord region of the Varanger Peninsula, eastern Finnmark, Norway are described.

2. General geology

The geology and stratigraphy of eastern Finnmark has been intensely studied during the last decades and is well understood (e.g., Siedlecka and Siedlecki, 1971; Siedlecki, 1980; Føyn, 1985; Rice et al., 1989 and references therein).

The Varangerfjord region forms the southern half of the Peninsula (Fig. 1). The bedrock consists of a succession of late Riphean and Vendian sediments divided into three units, which in ascending order are the Vadsø Group, the Tanafjord Group and the Vestertana Group. These units are separated from the allochthonous Barents Sea Region by the Trollfjord–Komagelv Fault Zone (Johnson et al., 1978). The Caledonides with the Kalak Nappe Complex covers the northwestern corner of the Peninsula (Fig. 1) and the Gaissa Nappe of the Lower Allochthon covers parts of the Peninsula in the Tanafjord area. The Vestertana Group is separated from the Tanafjord Group by a slight unconformity of 1 to 2° (Føyn, 1985). The formations and lithologies of the Vestertana and Tanafjord groups are given in Table 1.

In this study data are reported from the Stappogiedde and Nyborg formations of the Vestertana Group and from the Grasdal and Dakkovarre formations of the Tanafjord Group. Also included in this study are data from one site in the Dividalen area (D in Fig. 1, inset). The rocks sampled there belong to the autochthonous Dividal Group and are considered contemporary with the Stappogiedde Formation (Føyn, 1967). The sequence consists of sandstones, siltstones and shales.

Rock sequences along the northern shore of the Varangerfjord are little affected by tectonic deformation and are generally flat lying or have a

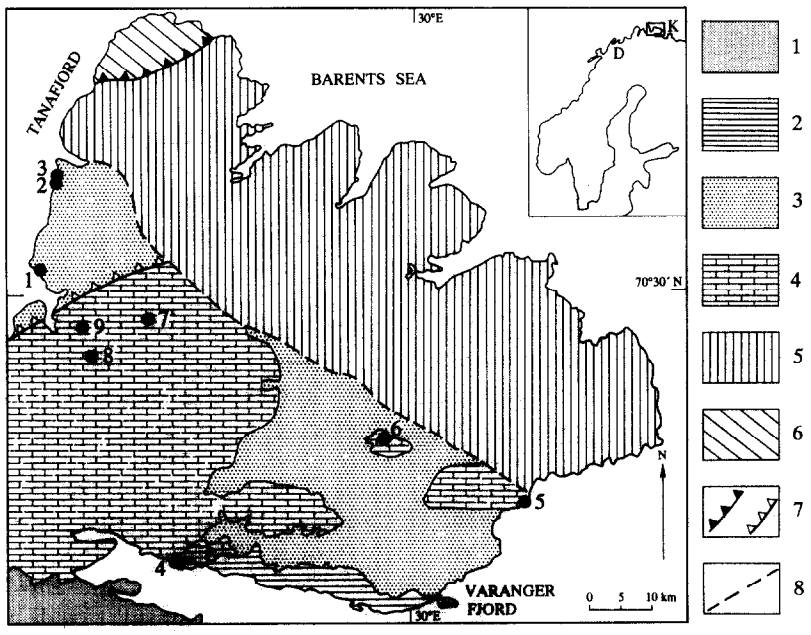


Fig. 1. Simplified geological map of the Varanger Peninsula. *1* = Precambrian basement; *2* = Vadsø Group *3* = Tanafjord Group; *4* = Vestertana Group; *5* = Barents Sea Region; *6* = Kalak Nappe *7*left = Caledonian Front of the Kalak Nappe, right = Caledonian Front of the Gaissa Nappe; *8* = Trollfjord–Komagelv fault. ● = palaeomagnetic sites. *D* and *K* denotes the Dividal and Kola Peninsula sites.

Table 1
Formations and lithologies of the Vestertana and Tanafjord groups (modified after Nystuen and Siedlecka, 1988)

Formation	Thickness (m)	Lithology
Vestertana Group		
Breivik Fm.	600	silty sandstone and shale
Stappogiedde Fm.	505–545	siltstone and sandstone
Mortensnes Fm.	10–60	glacial diamictite, mudstone
Nyborg Fm.	200–400	shale, siltstone, sandstone and dolomite
Smalfjord Fm.	2–50	glacial diamictite, conglomerate, siltstone and sandstone
Unconformity		
Tanafjord Group		
Grasdal Fm.	280	dolomite and shale
Hanglecærro Fm.	200	sandstone
Vagge Fm.	80	shale
Gamasfjell Fm.	280–300	sandstone
Dakkovarre Fm.	275–350	sandstone, siltstone and shale
Stangeness Fm.	205–255	shale and mudstone
Grønnes Fm.	130–200	conglomerate, sandstone and mudstone

shallow northerly dip. The incompetent Nyborg Formation, which is situated between two massive tillites, is folded in the eastern parts. Tectonic influence increases westward, and along the eastern shore of the Tanafjord in the Gaissa Nappe the strata are folded and the rocks are cleaved.

The Breivik Formation of the Vestertana Group has yielded body fossils and trace fossils that suggest an Early Cambrian age (Føyn and Glaessner, 1979) and they placed the transition between the Vendian and the Cambrian in the uppermost unit of the Stappogiedde Formation the Mandraperelv Member. Farmer et al. (1992) have described an Vendian Ediacaran assemblage from the central unit of the Stappogiedde Formation, the Innerelv Member.

The Nyborg Formation has yielded a Rb-Sr whole-rock age of 653 ± 23 Ma (Pringle, 1973) and the cleavage of the same formation has been dated to 635 Ma by the $^{40}Ar/^{39}Ar$ method on whole-rock samples (Dallmeyer and Reuter, 1989) and interpreted as the age of deformation. Dallmeyer et al. (1989) suggest that deformation and metamorphism of the Gaissa Nappe occurred in Late Ordovician–Early Silurian times, this based on K-Ar and $^{40}Ar/^{39}Ar$ ages. A K-Ar apparent age of 440.2 ± 9.4 Ma recorded by the < 0.5 μm size fraction isolated from metapelite from the Gaissa Nappe is interpreted to represent a maximum age for its metamorphism and deformation. The metamorphic conditions within the Gaissa Nappe are considered to be low-anchizone/intermediate-baric type to high-anchizone/high–intermediate-baric type with a corresponding temperature range between 210 and 290°C (Rice et al., 1989). The metamorphism within the autochthon decreases eastward.

3. Palaeomagnetic methods

Hand samples were collected and in-situ orientation was made by a magnetic compass and a sun compass. Sampling sites are shown in Fig. 1. From each sample up to five specimens were drilled and cut for measurements of the Natural Remanent Magnetization (NRM). These were made on Geofyzika's JR-3 and Jr-5 magnetometers and on Digico and Molspin fluxgate magnetometers. Stepwise Alternating Field (AF) and thermal demagnetization techniques were used for isolating the characteristic remanence components (chRM). Chemical demagnetization was applied to a few specimens by immersing them in 8 *N* hydrochloric acid for periods of up to 1800 hours (Collinson, 1967).

During the thermal demagnetization, changes

in magnetic mineralogy during heating were monitored by measurements of the magnetic susceptibility. The heating was performed in a Schonstedt TSD-1 thermal specimen demagnetizer. The field in the oven and cooling chamber was less than 20 nanotesla during heating and cooling.

4. Palaeomagnetic results

The measurements presented a mixed pattern of behaviours with stable and unstable magnetization. Data from about 50% of the samples had to be discarded either because their measured

Table 2
Site mean directions and Virtual Geomagnetic Poles (VGP)

Formation, site	N/n	Decl. (°)	Incl. (°)	α_{95} (°)	k (°N)	P lat. (°E)	P long. (°)	D_p (°)	D_m
Tanafjord Group									
Dakkovarre Fm.									
1 Vagge section	7/12	166	68	12	11	32	39	17	21
t.c.	7/12	159	−28	11	13	33	233	7	12
Grasdal Fm.									
2 Grasdal	5/7	155	47	10	31	10	51	8	12
t.c.	5/7	131	70	13	16	39	64	20	23
Vestertana Group									
Nyborg Fm.									
3 Grasdal	5/7	55	41	14	9	33	145	11	17
t.c.	5/7	41	16	14	10	22	164	11	17
4 Mortensnes	7/18	111	38	10	12	13	92	7	11
t.c.	7/18	104	46	11	10	21	96	9	14
Stappogiedde Fm.									
5 Komagnes A	10/14	59	64	6	37	53	130	8	10
t.c.	10/14	74	66	6	44	50	113	8	9
Komagnes B	5/10	172	19	14	10	−10	39	8	14
t.c.	5/10	172	8	15	9	−16	39	7	15
6 Bjornskardet	5/10	135	26	12	14	−1	73	7	13
t.c.	5/10	131	24	11	18	−1	78	6	11
7 Julahaugen A	6/7	84	46	11	23	28	115	9	14
t.c.	6/7	93	42	11	23	22	108	8	13
Julahaugen B	6/8	88	−19	14	13	−9	124	7	14
t.c.	6/8	86	−21	14	13	−10	126	8	15
8 Gæt'keg'årsa	5/9	255	69	17	8	−44	155	25	29
t.c.	5/9	246	68	18	8	−40	161	25	30
9 Juladalen	5/14	270	64	10	16	−42	139	12	15
t.c.	5/14	274	57	9	17	−36	131	10	13
Dividal Group									
10 Dividalen	8/11	63	67	11	16	55	110	14	17
t.c.	8/11	57	68	10	14	58	114	16	19
Mean sites 2, 4, 5A, 7A, 10	5	100	58	20	10	33	96	21	29
t.c.	5	93	60	14	20	37	101	16	21

N = number of samples and n = number of specimens from each site that gave characteristic directions; specimen means are presented in the table. Decl. and Incl. = site mean declination and inclination, respectively; α_{95} = the semi-angle of the 95% cone of confidence; k = best estimate of Fisher's (1953) precision parameter; P lat. = calculated latitude of the VGP, P long. = the calculated longitude of the VGP; D_p and D_m = the semi-axis about the VGP at the 95% probability level. t.c. = tilt corrected.

magnetic intensities were below the noise levels of the instruments used, or because the magnetic directions became erratic. In Table 2 are presented the number of samples (N) and specimens (n) that yielded chRMs. Fisher (1953) statistics were used for site mean calculations. Specimen means are given in Table 2. The characteristic components were obtained using a least-squares algorithm (Torsvik, 1986) combined with analysis of stereograms. Examples of the magnetic behaviour during demagnetization are given in Figs. 2–5. Site mean directions are given in Fig. 6.

4.1. The Dakkovarre Formation

This formation was sampled across the Vagge section on the eastern coast of the Tanafjord at site 1 (Fig. 1; Siedlecka and Siedlecki, 1971). Samples were taken in quartzitic and ferruginous sandstones. The strata are near-vertical and part of a complex fold system (Siedlecki, 1980). Southeastern directions with intermediate to steep positive inclination (Fig. 2b and c; Table 2) were isolated.

4.2. The Grasdal Formation

Samples from this formation were collected at outcrops along the shore at the type locality at Grasdal on the east side of the Tanafjord (site 2 in Fig. 1). They consist of fine-grained sandstone and dark siltstones from the lower parts of the formation and buff and grey dolomites from its upper part. The strata dip ca. 30° towards the northwest. A SE direction with intermediate positive inclination was isolated (Fig. 2a; Table 2).

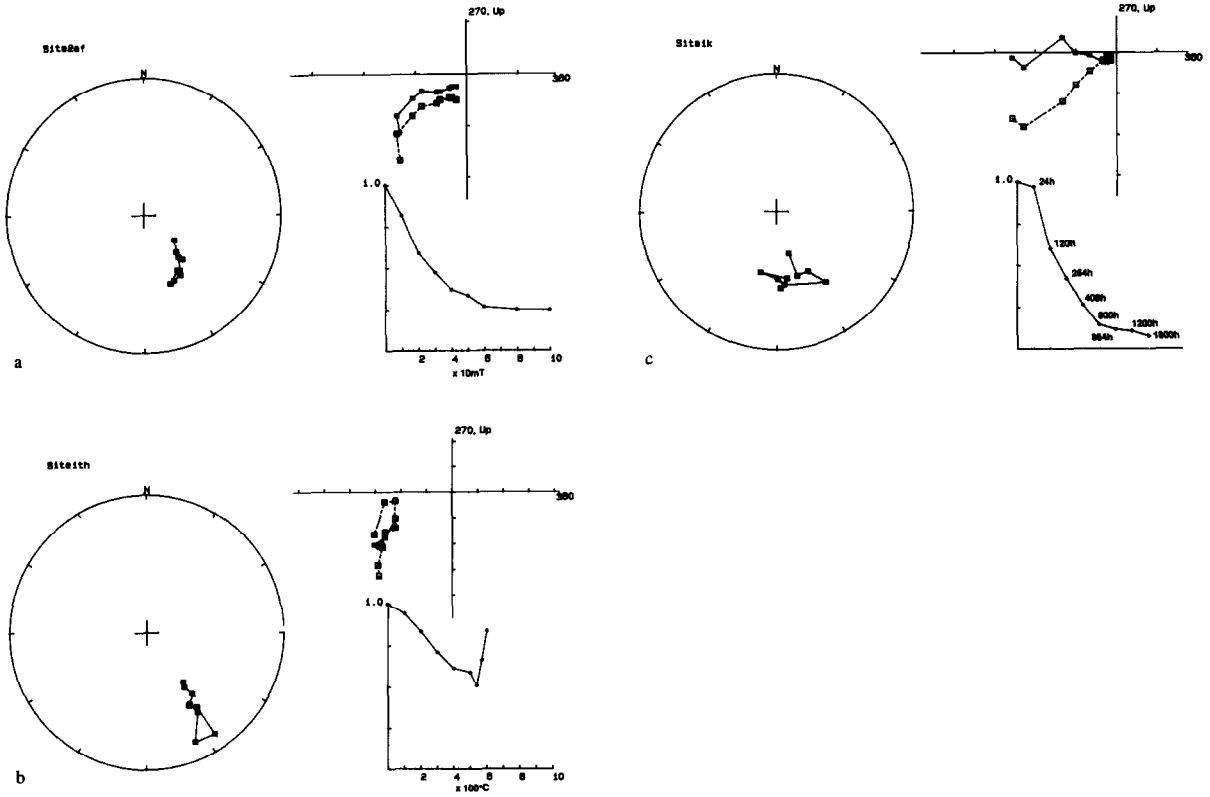


Fig. 2. Examples of behaviour of specimens from Dakkovarre and Grasdal Formations during AF (a), thermal (b) and chemical (c) demagnetization. In each figure the left part presents stereograms where closed (open) symbols denote positive (negative) inclination. Right top present orthogonal vector projections where open (closed) symbols denote points in the vertical (horizontal) plane. Right bottom presents normalized demagnetization curves. NRM_0 = 1.2 mA/m (a), 4.8 mA/m (b) and 3.0 mA/m (c).

The demagnetization of specimens from sites 1 and 2 indicate the presence of low-coercivity phases (Fig. 2). The thermal demagnetization yielded directions similar to these resolved by AF demagnetization up to temperatures of 500–540°C. At higher temperatures there was a marked increase in magnetic intensity (Fig. 2b). This was confirmed by bulk susceptibility measurements indicating the presence of iron phases, e.g., goethite, that become oxidized to magnetite during the heating process. The chemical demagnetization isolated the same characteristic components as the AF and thermal demagnetizations (Fig. 2c).

4.3. *The Nyborg Formation*

This locality is situated about 500 m north of site 2 (the Grasdal Formation) on the eastern shore of the Tanafjord (site 3 in Fig. 1). Grey, green and reddish laminated siltstones were collected. The samples are from the lower part of the formation ca. 5 m above the contact with the tillite of the Smalfjord Formation. The strata dip ca. 30° to the north-northwest. A northeastern component with intermediate positive inclination (Fig. 3a; Table 2) was obtained.

Site 4 is a cliff section above the northern shore of the Varangerfjord (Fig. 1) at Mortensnes. The samples were collected in red and grey sandstones close to the contact with the tillite of the Mortensnes Formation. The strata dip 10–15° to the north. One component with ESE direction with intermediate positive inclination was isolated (Fig. 3b and c; Table 2).

AF demagnetization up to 100 mT of samples from the Nyborg Formation, sites 3 and 4, reduced the NRM with only 20–50% (Fig. 3a and

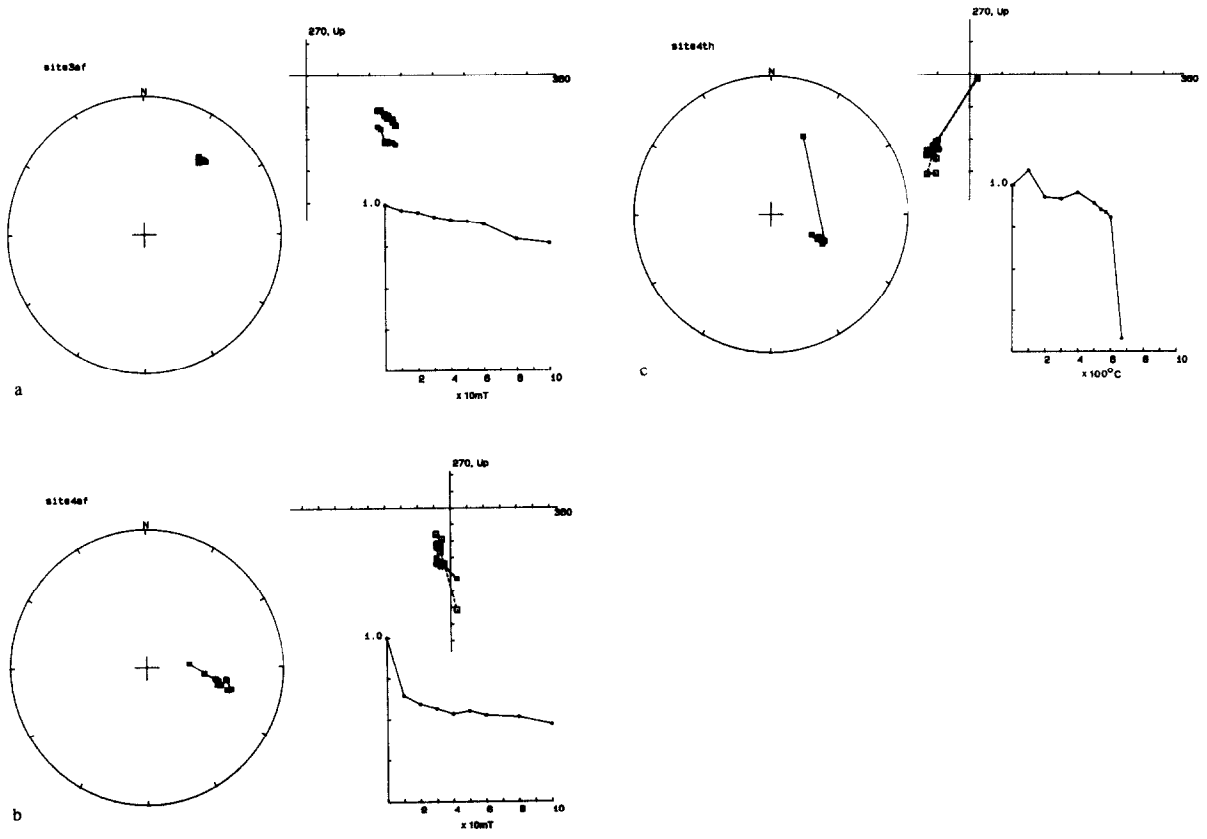


Fig. 3. Examples of behaviour of specimens from the Nyborg Formation during AF (a and b) and thermal (c) demagnetization. Conventions as in Fig. 2. NRM_0 = 3.9 mA/m (a), 8.6 mA/m (b) and 22.8 mA/m (c).

b). This together with the thermal demagnetization that showed a Curie temperature of ca. 670°C (Fig. 3c) indicates that haematite is the main carrier of the chRM in these rocks.

4.4. The Stappogiedde Formation

This formation is divided into three members, in ascending order the Lillevatn Member, the Innerelv Member and the Manndraperelv Member. For this study samples were collected in the Innerelv and Manndraperelv members. The Innerelv Member was collected at sites 5 (Komagnes) and 6 (Bjørnskardet) (Fig. 1). At site 5 the samples were collected in a cliff exposure on the shore of the Varangerfjord in grey to green siltstones and mudstones. The strata have a shallow dip towards the south-southwest. Two components, named A and B, were isolated in the samples of site 5. The A component has a NE trend with intermediate to steep positive inclination (Fig. 4a and b) and the B component has a positive shallow SE direction. There was no specimen where it was possible to isolate both components and no relationship was found between stratigraphical level and magnetic direction.

At Bjørnskardet (site 6) a shallow positive SE direction similar to the B direction found at site 5 was obtained. The samples consist of red and grey sandstones–siltstones. The strata dip ca. 10° to the northeast. The demagnetization tests imply a complex magnetic mineralogy with phases with

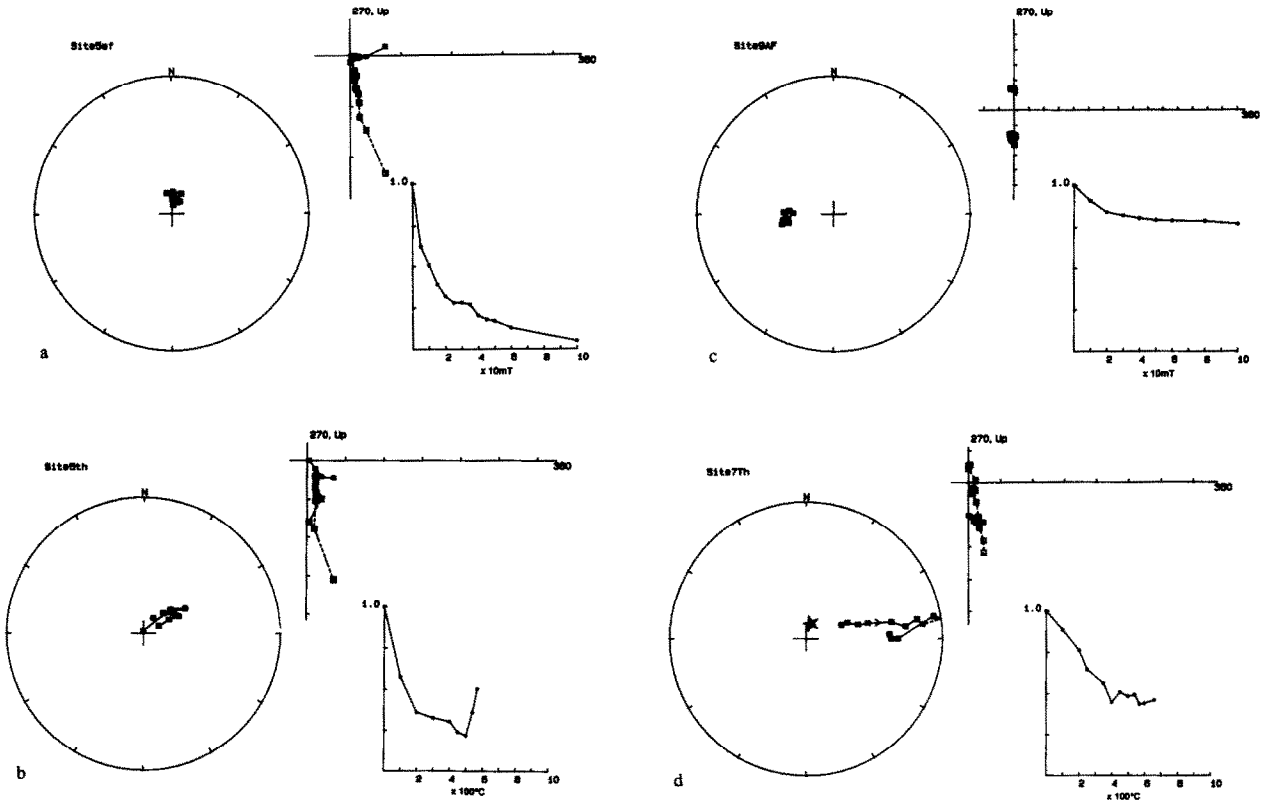


Fig. 4. Examples of behaviour of specimens from the Stappogiedde Formation during AF (a and c) and thermal (b and d) demagnetization. Conventions as in Fig. 2. ★ in (d) denotes the direction of the Earth's present field. NRM_0 = 28.8 mA/m (a), 16.0 mA/m (b), 1.6 mA/m (c) and 1.3 mA/m (d).

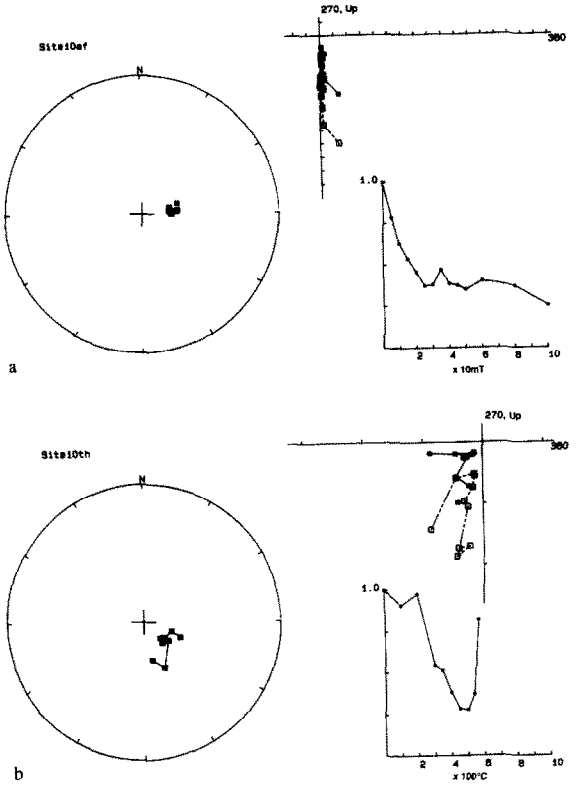


Fig. 5. Examples of behaviour of specimens from the Dividalen Group during AF (a) and thermal demagnetization (b). Conventions as in Fig. 2. $NRM_0 = 9.2$ mA/m (a) and 10.2 mA/m (b).

low to intermediate coercivity and blocking temperature.

The Manndraperelv Member was sampled at site 7 (Julahaugen); site 8 (Gæt'kegår'sa) and site 9 (Juladalen); all situated in the western part of the Peninsula (Fig. 1) and close to the front of the Gaissa Nappe. Red sandstones were collected at all three sites. The strata dip 5–10° to the southwest at sites 7 and 8 and ca. 10° to the west at site 9. At sites 8 and 9, one component with high coercivity (Fig. 4c) and high unblocking temperature was isolated in all samples indicating haematite as carrier of the remanence. At site 7, two components were isolated, 7A with a low blocking temperature pointing east with intermediate positive inclination followed by a component 7B with a high blocking temperature also pointing east, but with shallow negative inclination (Fig. 4d). The 7B direction may represent an incomplete reversal with respect to the directions found at sites 8 and 9.

4.5. The Dividal Group

At the Dividal Group, site 10 (68.8°N, 19.8°E; Fig. 1, inset; Table 2) were samples collected in red mudstones. A single component pointing east with positive intermediate to steep inclination

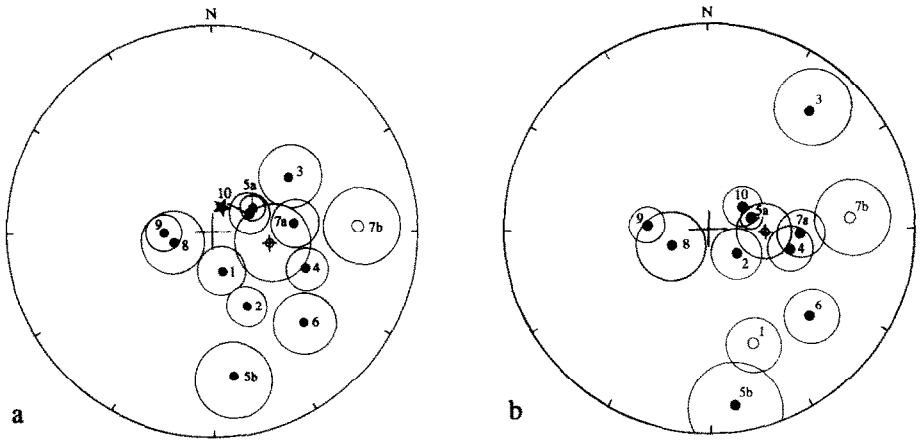


Fig. 6. Site mean directions after demagnetization. (a) Before tilt correction (b) After tilt correction. Circle with cross denotes the mean of site directions 2, 4, 5A, 7A and 10. ★ in (a) is the present direction of the Earth's field. ● (○) denote positive (negative) inclination.

was obtained. AF demagnetization showed a component with intermediate coercivity (Fig. 5a). During the thermal demagnetization the intensity increased above 450°C indicating mineralogical changes during heating (Fig. 5a and b).

5. Discussion

Table 2 and Fig. 6 present site mean directions before and after tilt adjustment to the palaeohorizontal. Table 3 and Fig. 7 give the corresponding palaeopoles together with some published Devonian to Late Riphean palaeopoles from Fennoscandia. The poles obtained in this study are from single sites and do not necessarily represent formations or longer time sequences; they are therefore considered as Virtual Geomagnetic Poles (VGP). In Fig. 7 both uncorrected and corrected VGPs are given joined by a line. An exception is the site 1 tilt-corrected VGP which is outside the figure. One reason for presenting both groups of poles is the tighter grouping of directions after tilt correction for sites 2, 4, 5A, 7A and 10 (Table 2) indicating a possible pretilt age for their magnetization although the improvement in k is not significant at the 95% confidence level and the shallow dips of the strata, generally less than 15°, result in only minor adjustments to the pole positions. An obvious exception is site 1 where bed-

Table 3
Palaeopoles used for comparison with the Tanafjord–Vestertana VPGs

Area, rock unit	S lat. (°N)	S long. (°N)	N	P lat. (°N)	P lon. (°E)	D_p (°)	D_m (°)	Age Ma	Ref.
1 Sredniy Peninsula sandstones, carbonates	69.5	32.0	186	53	82	5	6	620	1
12 Sredniy Peninsula dolerites	69.5	32.0	12	52	52	17	17	600	2
13 Kildinskaja Series grey beds	69.5	32.0	99	50	84	6	7	600	2
14 Kildinskaja Series	69.4	32.8	178	50	86	5	5	600	2
15 Kildin Island grey beds	69.3	34.0	67	49	94	8	9	600	2
16 Kildin Peninsula grey carbonates, terrig. rocks	69.3	34.0	61	51	98	8	9	600	1
17 Båtsfjord dykes	70.7	29.8	6	41	108	5	7	640	3
18 Nexö sandstone	54.2	15.3	14	38	134	7	11	600–550	4
19A Fen carbonate Complex	59.3	9.3	19	63	142	3	4	600–530	5
19B Fen Complex	59.3	9.3	20	50	144	5	8	600–530	6
20 Alnö Complex A	62.5	17.5	21	−8	92	5	9	589–545	7
21 Sørøy–Oksna–Altenes–Alta gabbro comb. comp.C	70	25	10	31	86	11	15	530–490	8
22 Askøy mafic pluton	60.7	5.6	12	−25	53	5	10	474–444	9
23 Swedish limestones	59.9	15.7	73	30	46	3	4	Ordov. 470	10
24 Västergötland limestones	58.4	13.8	55	14	49	5	6	m. Llanvirn.–l. Llandeil.	11
25 Västergötland limestones	58.4	13.8	24	5	34	5	7	l.Llandeil.–m.Caradoc.	11
26 Mosjøen gabbro	64.9	13.3	23	25	186	3	5	420–400	12
27 Sulitjelma gabbro	67.2	16.2	22	17	175	3	7	420–400	12
28 Råna norite	68.3	17.0	19	13	188	4	8	420–400	12
29 Mean Devonian	–	–	–	18	152	–	–	375	13
30 Mean Vadsø–Barents Sea groups	70.5	30.0	9	23	209	5	9	750–700	14

S lat. = site latitude, S long. = site longitude, respectively; N = number of sites, samples or specimens used for calculation of the palaeopoles. Otherwise as in Table 2.

References: 1 = Khramov, 1989; 2 = Khramov, 1986; 3 = Kjøde, 1980; 4 = Prasad and Sharma, 1978; 5 = Poorter, 1972; 6 = Piper, 1988; 7 = Piper, 1981; 8 = Torsvik et al., 1990; 9 = Rother et al., 1987; 10 = Claesson, 1978; 11 = Torsvik and Trench, 1991; 12 = Piper et al., 1990; 13 = Pesonen et al., 1989; 14 = Bylund, in press.

ding is vertical and the rocks form part of an complex fold structure, for this site the structural data are too few to use for reorientation of the remanence vector to the palaeohorizontal.

Two tentative parts of the Fennoscandian APWP are shown in Fig. 7, the first is late Riphean to late Vendian (ca. 850 to 600 Ma) and based on data from Elming et al. (1993) and Bylund (in press). The other part is tentative and starts with the Ordovician poles listed in Table 2 and ends with the mean Devonian pole of Pesonen et al. (1989). Torsvik et al. (1990) compared the pole from the Båtsfjord dykes (Table 3; Kjøde, 1980) with Ordovician palaeopoles from southern Sweden and suggested that the Båtsfjord dykes became remagnetized during the Early Ordovician Finnmarkian event of the Caledonian Orogeny. The 600 Ma part of the APWP is based on data from late Riphean–Vendian rock units from the Rybachiy–Sreidny Peninsula and Kildin island on the northern coast of the Kola Peninsula (K in inset, Fig. 1). Shipunov (1988), however, points out that these units (entries 13–16 in Table 3) may have a postfolding magnetization of the same age as intruding diabase dykes. Shipunov's (1988) data correlate with data from sites 3, 4, 5A, 7A and 10 of this paper. Sequences from the late Riphean–early Vendian Vadsø Group and the Barents Sea Group of the Varanger Peninsula (group B data from Bylund, in press) give palaeopoles in the 750–700 Ma range of the APWP. These rock units are from the central and eastern part of the Peninsula and the data suggest that this part was not affected by the Finnmarkian event. Further evidence for this is provided by the lack of thermal influence on microfossils found in some of the sequences (Vidal, 1981; Vidal and Siedlecka, 1983). However, the Båtsfjord dykes are situated closer to the Caledonian Front and may therefore have been affected by Caledonian events. From the present study sites 1, 2, 3, 7, 8 and 9 are from the western and the most tectonized part of the Peninsula. Of these site 1 and 2 show "Ordovician" in-situ directions while 5B, 6 and 7B are close to the Devonian pole.

Site 3, the Nyborg Formation, is situated close to the Grasdal Formation site 2 and both should

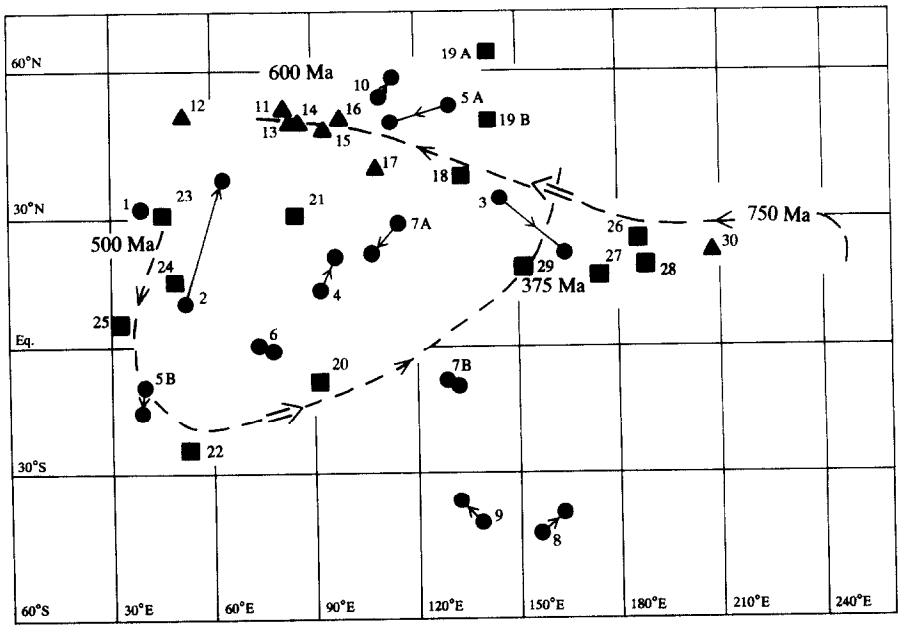


Fig. 7. VGPs and palaeopoles from Tables 2 and 3. The dotted lines denote tentative parts of the Fennoscandian APWP for 750–600 Ma and 500–375 Ma (Early Ordovician–Devonian). Polepositions from this paper (circles) are given both before and after tilt correction. They are joined by an arrow. ▲ denote Late Precambrian poles from Table 3 and ■ Palaeozoic poles.

have presented similar directions if remagnetized during the same event. However, there is a significant difference: decl. 155° and incl. 47° for the former; decl. 55° and incl. 41° for the latter. The AF-demagnetization tests (Figs. 2a and 3a) show that the chRM in the site 3 rocks is carried by haematite while the carrier of the chRM in site 2 is more complex. This may explain why a remagnetization event can have affected the rocks in different ways giving them different magnetic directions.

The Stappogiedde Formation components from sites 5A and 7A and the Dividal Group site 10 VGPs are situated in the vicinity of the 750–600 Ma range of the APWP together with poles from the Kola Peninsula and poles 18 and 19 from untectonized areas of Fennoscandia. The 7A direction is a low-temperature low-coercivity component (Fig. 4d; Table 2) and it is doubtful if it could have survived as a primary component, especially as the site is situated in the tectonically affected area close to the Gaissa Nappe where the red sandstones of the same site possess a hard magnetization carried by haematite.

Tectonic movements may have affected the positions of the VGPs, and could have caused the deviating positions of VGPs 8 and 9. The Båtsfjord dykes (pole 17, Table 3) are situated in the allochthonous Barents Sea Region north of the Trollfjord–Komagelv Fault. Bylund (in press) has shown that the amount of movement is less than 250 km by using the same rotation pole at 77°N, 60°E. that was used by Kjøde et al. (1978) for their reconstruction but based on new palaeomagnetic data from both sides of the fault. On geological grounds Rice and Townsend (1991) restricted the movement to 50 ± 50 km.

The gap in the APWP presented in Fig. 7 between the Vendian and the Ordovician parts is comparatively short, ca. 30°, indicating low movement rates for Fennoscandia. However, this period was a time signified by considerable movements between continents and the possible break up of a supercontinent (Hoffman, 1991 and references therein). This result suggests that rapid apparent polar wander may have taken place and that a complicated APWP exists for this time interval, but that we do not at present have enough reliable data to outline this part of the Fennoscandian APWP. An indication that this movement may have occurred is the large distance between the Early Cambrian poles

The inclination values obtained from the Nyborg Formation (sites 3 and 4, Table 1), indicate a low-latitude position for the inter-tillite formation at the time of magnetization, between 8° and 27°. This result is in agreement with Edwards and Føyn (1981) who suggest that the presence of dolomites in the upper part of the Nyborg Formation is an indication of a low-latitude deposition. The position of the mean Devonian pole (29), the gabbro palaeopoles (27, 28 and 29) of Piper et al. (1990) and the late Riphean–Vendian pole 30 (Bylund, in prep.) indicate an overlap in the APWP and this further complicates interpretation. The site 3 VGP may represent a primary Vendian position or a Late Silurian–Devonian overprint.

The Smalfjord, Mortensnes and Nyborg formations are the subject of a separate investigation.

6. Conclusions

The diverging directions, the situation of the sample area close to, or within, a nappe complex and the possibility of Caledonian remagnetizations make it difficult to assign ages to the obtained palaeopoles. However the results obtained indicate that the Innerelv Member of the Stappogiedde Formation at its easternmost site (site 5, Komagnes), may contain a component (5A) of primary origin. A contemporary unit (Dividal Group, site 10) ca. 400 km southwest of the Varanger Peninsula, similarly may retain a primary magnetization. This interpretation is based on comparison with data from Vendian–Early Cambrian units on the Kola Peninsula, the Nexö Sandstone and the alkaline intrusions of Fen and Alnö all isolated from Caledonian influence.

Directions indicating an Early Ordovician remagnetization were obtained from site 1 and 2 within the Gaissa Nappe, both before tilt correction. The Nyborg Formation site 3, also situated within the nappe, yields a direction that may

either be primary and Vendian, or represent a Silurian–Devonian remagnetization. The upper part of the Nyborg Formation at site 4 (Mortensnes), has an intermediate position between the two suggested APWPs and its significance is uncertain. Further, VGPs 5B, 6 and 7B are situated along a tentative Ordovician–Devonian part of the APWP. The discussion above assumes that little apparent polar wander occurred between Early Cambrian and Early Ordovician times. To clarify this problem it will be necessary to study late Vendian–Cambrian rocks from stable parts of the Fennoscandian Shield well away from Caledonian influences.

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